

Ziyu (Rick) Xu

Robotics Institute, Carnegie Mellon University

Research Interests: Robot Foundation Models, Real-World Policy Adaptation, and Learning-Augmented Control

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Education

Carnegie Mellon University (CMU)

Master of Science in Robotics (MSR)

Aug. 2026 – Expected 2028

Pittsburgh, PA

University of Michigan, Ann Arbor (UM)

B.S.E. in Mechanical Engineering (Robotics Track); Minor in Computer Science GPA: 3.85/4.0 (Top 5%)

Oct. 2024 – May 2026

- **Core Coursework:** Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization

Shanghai Jiao Tong University (SJTU)

B.S.E. in Electrical and Computer Engineering

Sep. 2022 – Jul. 2026

GPA: 3.56/4.0 (Top 20%)

- **Core Coursework:** Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems

Research Experiences and Internships

Intelligent Control Lab, Carnegie Mellon University

Graduate Student Researcher, advised by Prof. Changliu Liu

Aug. 2026 – Present

Pittsburgh, PA

- **Whole-Body and Safe Locomotion Research (ongoing):** Investigating whole-body control, safety with control barrier functions (CBFs), and world-model-based locomotion.
- **Dexmate Bimanual Manipulation Demo, IROS 2026:** Contributed to a bimanual block-stacking demonstration combining agent planning with policy execution. Implemented the agent component, prepared the on-site environment, adapted the system to different grippers, and performed system calibration.

Unitree Robotics, R&D Department, Embodied AI Team

Research Intern, Humanoid Robot Foundation Models

May 2026 – Present

Hangzhou, China

- **Closed-Loop Real-Robot Infrastructure for Humanoid Foundation-Model Policies:** Built and validated an end-to-end policy infrastructure on Unitree H2/G1 humanoids, integrating teleoperation data collection, VLA/WAM and whole-body policy adapters, rollout logging and evaluation, controller interfaces, and policy fine-tuning. Enabled launch with one command and closed-loop rollouts across platforms, allowing policies and controllers to be integrated without modifying platform-specific launch workflows.
- **Embodiment VLA Adaptation & Post-Training:** Conducted a structured survey of representative robot VLA, WAM, and physical-AI policies, especially work from NVIDIA GEAR Lab. Deployed and fine-tuned Isaac GR00T/OpenPI VLA policies with SONIC whole-body control on Unitree humanoids using real-robot rollout traces. Currently developing a phase-aware adaptation method that aligns semantic task progress and execution timing across embodiments for policy transfer and rollout-driven post-training.
- **Reproducible Real-World VLA Benchmark for Humanoid Manipulation:** Implemented an 8-task, 24-scene benchmark with green-screen background replacement to evaluate foundation models under varied visual conditions. Built a PICO-based scene-digitization and AR-guided reconstruction pipeline that captured shelf-object layouts as reusable digital scene templates, and registered target placements back into the physical workspace, achieving 2.5 cm median placement error.

Shanghai Artificial Intelligence Lab & IRMV, SJTU

Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang

Apr. 2026 – Present

Shanghai, China

- **Context-Aware Re-encoding for Iterative Visual KV Compression:** Identified representation errors caused by reusing retained visual KV entries after their spatial neighbors are removed, and showed that these errors compound across repeated compression in multi-turn and streaming inference. Developed a re-encoding method that recomputes each retained set in its new context at the original positions, and benchmarked it against inheritance and restoration methods under matched memory and runtime budgets.

Computational Autonomy and Robotics Lab, UM

Research Assistant, Directed by Prof. Maani Ghaffari

Jun. 2025 – Apr. 2026

Ann Arbor, MI

- **Online Learning Lie-MPC for Robust ASV Control:** Developed a Neural-Fly-inspired learning-augmented Lie-MPC framework for an autonomous surface vehicle, estimating residual wind/current disturbances online and injecting learned dynamics corrections into the controller for robust trajectory tracking under hydrodynamic uncertainty. Implemented and evaluated the framework on a real ASV, achieved a 20% improvement in trajectory-tracking performance over the baseline MPC controller during field experiments.

- **Field Perception for Aquatic Vegetation Tracking:** Built a water-scene perception pipeline for autonomous surface vehicles, combining DINO/SAM2-based segmentation with Grounded-DINO and YOLO fine-tuning to detect submerged vegetation and provide perception signals for navigation and autonomy experiments.
- **ASV Hardware and Sensor Integration for Closed-Loop Field Autonomy:** Designed sensor mounts and integrated GPS, lidar, IMU, and camera systems on an autonomous surface vehicle. Aligned and calibrated sensors to support field data collection and control experiments.

Intelligent Robotics and Autonomy Lab , UM

Oct. 2024 – Apr. 2026

Research Assistant, Directed by Prof. Vasileios Tzoumas

Ann Arbor, MI

- **Multi-Robot Active-SLAM Simulation & ROS2 Integration:** Rebuilt a Unity + AirSim multi-drone simulation testbed with redesigned vehicle dynamics and sensing modules, then migrated SlideSLAM communication logics from ROS1 to ROS2 by refactoring topic interfaces and workflows for bandwidth-limited active-SLAM evaluation.
- **Resource-Aware Multi-Robot Exploration Algorithms:** Developed distributed exploration EXP3-style algorithms for bandwidth-limited robot teams, formulating communication-aware action selection with submodular objectives and bandit feedback in simulation to balance map coverage and communication cost.

Surgical Computer Vision Detection, SJTU

Apr. 2024 – Dec. 2024

Research Assistant, Directed by Prof. Yutong Ban

Shanghai, China

- **Foundation Segmentation Model Evaluation for Surgical Video:** Evaluated Segment Anything Model variants on surgical video frames, analyzing domain shift and temporal mask stability around instruments, tissues, and tool-tissue interaction regions.

Publications and Manuscripts

- Xiaohan Wang*, **Ziyu Xu***, Xuyi Yang.
“Re-encoding What You Keep After Every Compression: Retained Visual KV Is Computed in a Context That No Longer Exists”,
*Submitted to ICLR 2027. *Equal contribution.*
- Xuyi Yang, Xiaohan Wang, Wenhao Zhang, and **Ziyu Xu**.
“StreamJIT: Continuous Observation and Just-in-Time Deliberation for Proactive Streaming Video Assistants”,
Submitted to ICLR 2027.
- Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari.
“Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control”,
IEEE Robotics and Automation Letters (RA-L), revised manuscript under review, 2025.
- C. Yuan, J. Jiang, K. Yang, **Z. Xu**, . . . , and Y. Ban.
“Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis”, *npj Digital Surgery*, accepted, 2024.

Awards

- Gold Medal in 2023 University Physics Competition, by American Physical Society & American Astronomical Society (top 1%)
- Third place in 2023 RoboMaster Campus Competition (top 10%)
- Dean’s Honor List, University of Michigan College of Engineering (2024, 2025)
- M Prize in 2024 Mathematical Contest in Modeling (top 13%)

Skills

- **Robot Learning & Control:** VLA post-training, humanoid foundation models, whole-body policy, RL, real-robot policy deployment, teleoperation/data collection, MPC, sim-to-real
- **Perception & Multimodal ML:** PyTorch, OpenCV, SAM2, Grounded-DINO, YOLO, DINO features, visual token compression, spatial grounding, LLMs fine-tuning
- **Systems & Tools:** ROS/ROS2, Python/C++, Unity + AirSim, Docker, Linux, Git, sensor integration, CAD, FEA, Manufacturing, embedded system